Phylogenetic topology (Open Tree of Life)
Species / Taxon
Reported compounds (log scale)
0
1
10
50
100
165
Bulbophyllum flaviflorum
Bulbophyllum umbellatum
Bulbophyllum triste 4
Bulbophyllum taeniophyllum 2
Bulbophyllum lepidum
Bulbophyllum crabro 12
Bulbophyllum pumilum
Bulbophyllum japonicum
Bulbophyllum shweliense
Bulbophyllum auricomum 9
Bulbophyllum calyptratum
Bulbophyllum coweniorum
Bulbophyllum fuscopurpureum 3
Bulbophyllum acutiflorum
Bulbophyllum protractum 11
Bulbophyllum laxiflorum
Bulbophyllum wendlandianum 23
Bulbophyllum drymoglossum
Bulbophyllum griffithii
Bulbophyllum kaitiense
Bulbophyllum sterile
Bulbophyllum retusiusculum 27
Bulbophyllum ipanemense 3
Bulbophyllum elatum 1
Bulbophyllum cariniflorum
Bulbophyllum cylindraceum
Bulbophyllum echinolabium 45
Bulbophyllum rufinum
Bulbophyllum emarginatum 10
Bulbophyllum guttulatum 2
Bulbophyllum leopardinum 1
Bulbophyllum cheiri 6
Bulbophyllum inconspicuum
Bulbophyllum careyanum
Bulbophyllum maximum
Bulbophyllum gymnopus 5
Bulbophyllum pectinatum
Bulbophyllum psittacoglossum
Bulbophyllum reptans 14
Bulbophyllum odoratissimum 16
Bulbophyllum barbigerum
Bulbophyllum falcatum
Bulbophyllum lobbii
Bulbophyllum mutabile
Bulbophyllum involutum 3
Bulbophyllum ambrosia
Bulbophyllum scaberulum
Bulbophyllum intertextum
Bulbophyllum lupulinum
Bulbophyllum andersonii 8
Bulbophyllum levinei
Bulbophyllum variegatum 19
Bulbophyllum affine
Bulbophyllum weddellii 2
Bulbophyllum vaginatum 29
Bulbophyllum sulcatum
Bulbophyllum lilacinum
Bulbophyllum kwangtungense 7
Dendrobium chryseum 12
Dendrobium nobile 165
Dendrobium catenatum 3
Dendrobium crystallinum 9
Dendrobium loddigesii 37
Dendrobium aphyllum 15
Dendrobium polyanthum 11
Dendrobium crepidatum 5
Dendrobium chrysanthum 19
Dendrobium chrysotoxum 26
Dendrobium capillipes 7
Dendrobium fimbriatum 28
Dendrobium hancockii
Dendrobium secundum 5
Dendrobium fugax
Dendrobium salaccense
Dendrobium officinale 138
Dendrobium discolor
Dendrobium plicatile 26
Dendrobium fuscescens 2
Dendrobium affine
Dendrobium teretifolium
Dendrobium lowii 1
Dendrobium cumulatum 1
Dendrobium signatum 3
Dendrobium pachyglossum 5
Dendrobium xantholeucum 16
Dendrobium hainanense 10
Dendrobium rotundatum 5
Dendrobium hymenanthum
Dendrobium purpureum
Dendrobium planibulbe
Dendrobium subulatum
Dendrobium nodosum
Dendrobium parishii 7
Dendrobium crumenatum 18
Dendrobium moschatum 2
Dendrobium lindleyi
Dendrobium jenkinsii
Dendrobium linawianum 1
Dendrobium scabrilingue 7
Dendrobium pendulum
Dendrobium sinense
Dendrobium terminale 1
Dendrobium strongylanthum
Dendrobium monticola
Dendrobium wardianum 10
Dendrobium macraei
Dendrobium candidum 26
Dendrobium formosum 12
Dendrobium aurantiacum 7
Dendrobium lohohense
Dendrobium williamsonii 7
Dendrobium primulinum 1
Dendrobium comatum 2
Dendrobium trigonopus 8
Dendrobium amoenum 11
Dendrobium gibsonii
Dendrobium eriiflorum
Dendrobium denudans
Dendrobium canaliculatum
Dendrobium hookerianum
Dendrobium thyrsiflorum 20
Dendrobium farmeri
Dendrobium amplum 11
Dendrobium henryi
Dendrobium densiflorum 25
Dendrobium falconeri 7
Dendrobium senile 8
Dendrobium huoshanense 10
Dendrobium lituiflorum 1
Dendrobium denneanum 12
Dendrobium devonianum 10
Dendrobium aduncum 1
Dendrobium gratiosissimum 23
Dendrobium moniliforme 57
Dendrobium cariniferum 5
Dendrobium ellipsophyllum
Dendrobium brymerianum 11
Dendrobium findlayanum 4
Dendrobium ochreatum 1
Dendrobium ovatum
Dendrobium infundibulum 9
Dendrobium herbaceum 2
Dendrobium macrostachyum
Dendrobium tosaense 4
Dendrobium longicornu 33
Dendrobium draconis 5
Dendrobium bellatulum
Dendrobium hercoglossum
Liparis viridiflora
Liparis condylobulbon 4
Bulbophyllum caespitosum
Liparis nakaharai 8
Liparis cathcartii
Liparis petiolata
Liparis fargesii
Liparis rostrata
Liparis bootanensis
Liparis elliptica
Liparis stricklandiana
Liparis nigra
Liparis inaperta
Liparis dunnii
Liparis odorata
Liparis tschangii
Liparis nakaharae 8
Liparis formosana 1
Liparis distans
Liparis nervosa
Liparis cordifolia 2
Cymbidium kanran 1
Cymbidium ensifolium
Cymbidium sinense
Cymbidium goeringii 3
Cymbidium faberi
Cymbidium lancifolium
Cymbidium floribundum
Cymbidium devonianum
Cymbidium hookerianum
Cymbidium lowianum
Cymbidium eburneum
Cymbidium iridioides 4
Cymbidium dayanum
Cymbidium aloifolium 15
Cymbidium mannii
Cymbidium macrorhizon
Cymbidium bicolor
Cymbidium giganteum 4
Cymbidium wilsonii
Eulophia spectabilis
Eulophia petersii 1
Eulophia bicallosa 1
Eulophia ovalis
Eulophia ochreata 1
Eulophia epidendraea
Eulophia pratensis
Eulophia dabia
Eulophia tenella
Eulophia herbacea
Eulophia cucullata
Eulophia speciosa
Eulophia macrobulbon 4
Eulophia graminea
Peristylus jinchuanicus
Peristylus constrictus
Peristylus tentaculatus
Peristylus goodyeroides
Peristylus bulleyi
Peristylus affinis
Peristylus lawii
Peristylus densus
Calanthe triplicata 2
Calanthe calanthoides
Calanthe tricarinata
Calanthe arcuata
Calanthe discolor 11
Calanthe alpina
Calanthe cardioglossa 15
Calanthe ceciliae
Calanthe plantaginea
Calanthe puberula
Calanthe clavata
Calanthe brevicornu
Calanthe mannii
Calanthe lyroglossa
Calanthe odora
Calanthe hancockii
Calanthe whiteana
Calanthe densiflora
Calanthe graciliflora
Calanthe arisanensis 17
Calanthe alismifolia
Calanthe reflexa
Calanthe davidii
Calanthe vestita
Calanthe masuca 1
Eulophia ensata
Coelogyne fimbriata
Coelogyne stricta 5
Coelogyne barbata 3
Coelogyne chinensis 38
Coelogyne alba
Coelogyne articulata 5
Coelogyne cantonensis
Coelogyne cristata 4
Coelogyne flaccida 8
Coelogyne fuscescens
Coelogyne viscosa
Coelogyne elata 5
Coelogyne calcicola
Coelogyne candoonensis
Coelogyne ovalis 9
Coelogyne leucantha
Coelogyne sanderae
Coelogyne odoratissima 1
Coelogyne prolifera 1
Coelogyne punctulata
Coelogyne corymbosa 2
Coelogyne nitida 13
Coelogyne occultata
Pleione humilis
Pleione hookeriana
Pleione scopulorum
Pleione formosana 12
Pleione bulbocodioides 37
Pleione yunnanensis 15
Pleione praecox
Pleione maculata
Bletilla foliosa
Bletilla formosana 57
Bletilla striata 95
Bletilla ochracea 28
Arundina graminifolia 68
Nervilia aragoana 11
Gastrodia elata 63
Nervilia macroglossa
Nervilia plicata 33
Nervilia fordii 13
Habenaria pantlingiana
Habenaria repens 3
Habenaria glaucifolia
Habenaria marginata
Habenaria commelinifolia
Habenaria ciliolaris
Habenaria plantaginea 2
Habenaria pectinata
Habenaria limprichtii
Habenaria hollandiana
Habenaria rhodocheila
Habenaria diplonema
Habenaria digitata 5
Habenaria aitchisonii
Habenaria mairei
Habenaria floribunda
Habenaria diphylla
Habenaria fordii
Habenaria rariflora
Habenaria davidii
Habenaria linguella
Habenaria acuminata
Habenaria petalodes 6
Habenaria frappieri
Habenaria epipactidea
Habenaria crinifera
Habenaria petelotii
Habenaria roxburghii
Habenaria longicorniculata
Habenaria intermedia 3
Habenaria furcifera
Habenaria dentata
Habenaria delavayi
Habenaria tibetica
Dactylorhiza saccifera
Dactylorhiza majalis 2
Dactylorhiza maculata 4
Dactylorhiza incarnata
Dactylorhiza viridis
Orchis militaris 6
Orchis anthropophora
Orchis italica
Dactylorhiza hatagirea 17
Dactylorhiza iberica
Gymnadenia conopsea 85
Dactylorhiza romana
Dactylorhiza osmanica
Dactylorhiza umbrosa
Dactylorhiza kalopissii
Dactylorhiza baumanniana
Dactylorhiza urvilleana
Dactylorhiza durandii 2
Dactylorhiza cordigera
Gymnadenia orchidis
Gymnadenia rhellicani
Gymnadenia crassinervis
Dactylorhiza purpurella
Orchis mascula
Orchis chusua
Orchis punctulata
Orchis pallens
Orchis anatolica
Orchis spitzelii
Orchis simia
Orchis purpurea
Orchis provincialis
Orchis palustris
Peristylus forceps
Peristylus coeloceras
Cypripedium parviflorum
Cypripedium calceolus 1
Cypripedium macranthos 2
Cypripedium japonicum
Cypripedium guttatum
Cypripedium debile
Cypripedium micranthum
Cypripedium acaule 7
Cypripedium elegans
Cypripedium henryi
Cypripedium fargesii
Cypripedium margaritaceum
Cypripedium flavum
Cypripedium fasciolatum
Cypripedium tibeticum
Cypripedium bardolphianum
Cypripedium franchetii
Vanilla somae
Vanilla walkeriae
Vanilla griffithii
Vanilla madagascariensis 52
Vanilla dilloniana
Vanilla pompona 5
Vanilla mexicana
Vanilla planifolia 97
0
1
10
50
100
165
Compound pathway & superclass (5 shades per pathway row)
SHIKIMATES & PHENYLPROPANOIDS : Phenanthrenoids, Other superclass, Stilbenoids, Flavonoids, Phenolic acids (C6-C1)
TERPENOIDS : Sesquiterpenoids, Steroids, Triterpenoids, Other superclass, Monoterpenoids
FATTY ACIDS : Fatty acyls, Fatty esters, Fatty Acids and Conjugates, Other superclass, Sphingolipids
POLYKETIDES : Polycyclic aromatic polyketides, Naphthalenes, Cyclic polyketides, Other superclass, Chromanes
ALKALOIDS : Pseudoalkaloids, Tryptophan alkaloids, Other superclass, Ornithine alkaloids, Anthranilic acid alkaloids
CARBOHYDRATES : Saccharides, Nucleosides, Other superclass, Polyols, Aminosugars and aminoglycosides
AMINO ACIDS & PEPTIDES : Small peptides, Phenylpropanoids (C6-C3), Other superclass, Peptide alkaloids, Fatty Acids and Conjugates
OTHER : Other superclass
Known ethnobotanical species
$10^{-0.2}$ $10^{-0.15}$ $10^{-0.1}$ $10^{-0.05}$ 10^{0}
Known species
10^{0} 10^{1} 10^{2} 10^{3}